

SIMULTANEOUS *Go* VIA OBJECTIVE QUANTUM REDUCTION

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ABSTRACT. We construct a simultaneous combinatorial game *SGo* that is in a precise mathematical sense a simple symmetric simultaneization of the classical board game *Go*, where simplicity too is a precise mathematical notion. We develop some general theory of this symmetric simultaneization here, in particular on the universal level, and this may be of independent interest. Based on this we prove a theorem that chess and *Go* are qualitatively different in this theory: in the language here, the absolute radius (an isomorphism invariant) of *Go* is infinite, of chess zero. *SGo* is not just theoretical: it is a very playable game that is in some ways a simpler game than *Go*, as Komi, Ko and suicide rules are removed. On the other hand it has similar and ostensibly sharper dynamics, a fact we verify via computer simulation. Using the Nash equilibrium theorem, we show that a finite symmetric simultaneous combinatorial game has a mixed equilibrium strategy that draws on average, and so *SGo* does as well. The construction of *SGo* is inspired by physical quantum theory, despite the game being combinatorial.

1. INTRODUCTION

Go is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. However, this structure also has drawbacks: we need an ad hoc rule such as Komi to deal with the advantage of Black having the first turn. Moreover, the Ko situation and its cousins like triple Ko necessitate certain non repetition rules. Although non repetition rules are necessary in the end game of many types of games, like chess for example, in *Go* they are necessary even in the early game. This leads to Ko fights, in which the game loses its local-to-global character¹.

At the same time, partly due to its local-to-global nature, *Go* is almost simultaneous. We make this precise by proving that *Go* has simple (in a certain mathematical sense) symmetric simultaneizations of faithfulness radius any $k \in \mathbb{N}$ or ∞ , see Theorem 5.2. The radius zero game *SGo* is the star here, it is very playable with razor sharp dynamics². In contrast, we show that no simple simultaneization of suitably formalized chess has positive faithfulness radius, see Theorem 5.3.

SGo is a combinatorial symmetric simultaneous game, which is playable on a standard *Go* board, although we need a third type of stone that we call q-stones. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules: stone placement and capture (aside from the end of game rule). Its game dynamics are very similar to the classical *Go*; we make this precise

¹In the sense that if a player engages in a Ko fight, they must make moves that are strategically disconnected.

²The faithfulness radius k games we construct have progressively worse dynamics.

via some empirical studies outlined in Section 2.1. For example, on the 19×19 board, random games of *SGo* appear to be at least as sharp as those of *Go* (in fact the expected absolute game margin is much higher, but this must be properly interpreted). Furthermore, random *SGo* games at least empirically self-terminate (398 mean turns at 19×19 , with no non repetition rules invoked, in every one of a thousand games); in contrast, half of the *Go* games were still running after 1500 turns.³

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal simultaneization of certain sequential games developed in Section 4.4, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics, cf. Section 2.1. To get interesting dynamics in the case of *SGo*, we also have “objective reduction of the quantum state” – it is a deterministic, partial collapse of branches based on objective board information. The term objective reduction is coined by Penrose, see for instance [15]. Accordingly, a position of *SGo* carries its set of branches with it, as a position of chess carries its castling rights, see Section 2. Of course, the word quantum is used figuratively: there is no unitary time evolution, no complex amplitudes, and no probabilistic outcomes. The game is combinatorial and deterministic, and it would not be appropriate to call it “quantum Go”. For games with genuine quantum features see for instance [6], and for a quantum version of *Go* itself [17].

Using the Nash equilibrium theorem [13], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said, finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw, not just to draw on average; on the 2×2 board both players do, but already on the 3×3 board neither player has one, so optimal play is necessarily mixed (Section 4.2).

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [2], [19]. It also led to Conway’s theory of surreal numbers [4]. There is also research on simultaneous and synchronized combinatorial games, see for instance [3], [9], [10]. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

2. GAME RULES

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other’s move on this turn. We allow pass moves as in *Go*. The board the players see and play on we call the **display**; it is a **summary** of the game history, in a sense made precise in Section 4.1. Simultaneous action requires modifying the placement and capture rules of *Go*, while removing the Komi, Ko, and suicide rules. We also need to introduce a third type of stone, called **q-stones** (quantum stones), which may be

³These results come with caveats, a major one being that our rules for both games allow suicide, which is not standard. Furthermore, informed play is of course expected to produce vastly shorter games, for *Go* especially.

thought of as simultaneously white and black. On the board a q-stone is displayed *red* or, as a visual aid, *blue* when a capture is currently blocked through it, in the sense of the capture conditions below; the colors red and blue are purely visual and carry no rule content.

2.1. Motivation for the rules. Before we proceed to the main rules, we briefly overview the motivation and constraints we impose. We require:

- (1) *SGo* forms a simple simultaneization of *Go*, in the precise sense of Theorem 5.2. Being a simultaneization loosely means that *SGo* naturally maps to the universal simultaneization of *Go*.
- (2) It is playable on a standard *Go* board, which in particular requires us to represent simultaneous moves to the same position, and we choose to do this by introducing q-stones (visually red or blue).
- (3) We want the minimum amount of rules, in particular we do not want ad hoc rules dealing with Ko type situations.
- (4) We want *SGo* to have similar game dynamics to the standard *Go*, that is with nearly optimal play the positions should tend to become very sharp. This is far from the case for the universal simultaneization $\text{Sim}(\text{Go})$ of *Go*, as considered in Section 4.4. For $\text{Sim}(\text{Go})$, games tend to be drawn due to players lacking forcing options.

To achieve these dynamics, we want to “optimally” prune branched histories resulting from simultaneous action, using objective global board information. This is formalized by the capture resolution rule. This has the effect of giving players very forcing options, see Section 3.2 for some examples. But this is not without constraints: we need to prune branches so that the game stays symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 2.4. We also cannot prune too much, for example in the suicidal capture condition in Definition 2.2 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 2.3.

Requirement (4) is supported by computer experiments.⁴ The main comparison plays both games on their common rule base – the variant of *Go* fixed before Theorem 5.2 (suicide legal, no superko rule) – under uniformly random play over all legal board moves, passing only when none is available; 1000 games per variant (200 in the last column). On the 5×5 board the game bound of Section 4.2 is set to 40 turns; on 19×19 , *SGo* is left unbounded, while *Go* is stopped at 1500 turns:

⁴The simulations and exact computations reported in this paper, together with extensive machine verification of its rules, examples, and arguments – including an exhaustive census of the one-turn faithfulness conditions over every legal classical position of the 3×3 board – were carried out by Claude, an AI system by Anthropic.

	<i>SGo</i> 5×5	<i>Go</i> 5×5	<i>SGo</i> 19×19	<i>Go</i> 19×19
mean margin	13.6	12.0	235.3	68.8
draws	4.6%	5.4%	0.0%	4.5%
games ending on their own	99%	0%	100%	50%
mean length in turns	22.4	39.9	398	1106
q-stones created per game	3.3	–	8.6	–
q-stones on the final display	2.3	–	5.0	–
peak number of branches	4.2	–	8.7	–
final number of branches	3.4	–	3.7	–

Here the mean margin is the absolute difference of the two area scores, and a game ends on its own when the end of the game rule fires before the bound or cap (the 19×19 figures for *Go* are censored at the cap, so its true mean length is larger still). On 5×5 the margins are nearly identical and the draw rates similar, and the difference is termination: 99% of *SGo* games ended of their own accord within the bound – and with the bound lifted, every one of them does, in 22.4 turns on average – while every classical game was still churning when the bound stopped it. On 19×19 the difference is dramatic on both counts: random *SGo* ends by itself on a decisively settled board, with mean margin exceeding half the board, while half of the classical games were still cycling, far from settled, when stopped (every one of the thousand *SGo* games ended on its own).

Disabling the capture resolution – approximating the dynamics of the universal simultaneization of Section 4.4 – collapses the game completely: no stone is then ever removed, the two colors necessarily place equal numbers of stones, and all 500 simulated 5×5 games ended in exact draws. The capture resolution is the entire engine of the dynamics; the placement rule is dictated by the entanglement semantics.

One may wonder how informed play changes the above distributions; we partially address this in Section 6.1.

Finally, the last four rows of the table show that the entanglement stays small: a finished *SGo* game displays on average about two q-stones on the small board, and five among the 361 points of the large one, so the display stays nearly classical. The branch counts are obtained by tracking the branch sets consistent with the displays – an upper bound for the entanglement state (all 1000 of the 19×19 games were so tracked). At both sizes the final branch count stays of order one, essentially independent of board size (3.4 against 3.7) – a handful of branches on boards whose a priori branching is astronomical. For sharpness under optimal play, see the discussion of drawing strategies in Section 4.2.

2.2. Some definitions. Given a board state S , that is any (at the moment no legality restrictions) collection of white, black, q-stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A **unicolor component** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C **black**, respectively

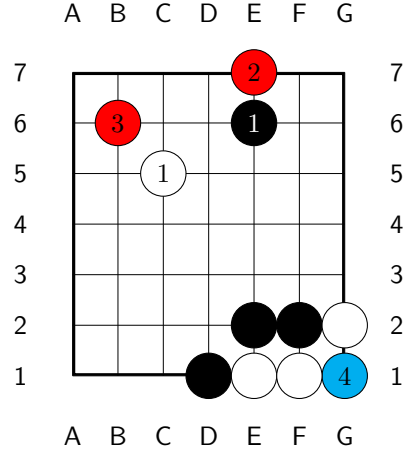
white. We may less often refer to **q-components**, analogously defined as connected components of q-stones.

As in classical Go , a connected component C **has no liberties** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a **trapped component**.

We define stones **adjacent** to C as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

2.3. Stone placement rule of SGo . If the players move to the same intersection i on the board, a q-stone is placed at i .⁵

To the right is an example of the placement rule. On move one the players move to different intersections and both stones are placed as in classical Go . On moves two, three and four they move to the same intersection – E7, B6 and G1 respectively – and each conflict yields a q-stone. (One can check that no capture then occurs on turn 4: the blue stone blocks both parts of condition 3 for the white component E1, F1.)



2.4. Capture rule of SGo . After both White and Black move on the current turn numbered n , using the stone placement rule above we get some board state S .

Definition 2.1. We say that a black, white, or q-component C is **trapped on turn n** if C is trapped and n is the last turn on which a stone was placed in or adjacent to C . A stone is called **trapped** on turn n , if it is an element of C as above. We say that a q-stone is **trapped by white respectively black on turn n** (white, respectively black, being the capturing side), if replacing it by a black respectively white stone it is trapped on turn n . For a black respectively white stone, trapped by white respectively black on turn n will simply mean trapped on turn n .

Definition 2.2. A white component C is resolved as **captured on this turn** if the following is satisfied:

- (1) C is trapped on this turn.
- (2) If C is adjacent to a q-stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 2.4 does not work, and the game evolution will be ill defined.)
- (3) One of the following holds:

⁵In terms of the entanglement state: each branch gains the stone of one of the two players (in each serialization the second move to i is forfeited), and all branches are kept. The branches then disagree at i , so displaying either black or white would misrepresent half of the entanglement: a q-stone is the faithful summary, and the rule is forced.

- (a) No black stone or q-stone adjacent to C is trapped by white on this turn.
- (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then the component of that stone contains a stone that is created on this turn and is not a q-stone.

Remark 2.3. *The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.1. For example, if we remove the part of the condition “is not a q-stone”, then Ko type infinite loops will become possible later. If we remove the condition “no stone in C is created on this turn”, then Theorem 5.2 will fail.*

In the case C is captured:

- The stones of C are removed as prisoners of Black as in *Go*.
- Any q-stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

A stone converted this way is not considered to be placed or created on this turn; it retains the placement turn of the q-stone it replaces.

If C is a black component the process is naturally mirrored. As far as the rules are concerned this process is simultaneous for all captured components, cf. Lemma 2.4.

2.5. Stage one reduction. The above capture procedure is called **stage one reduction** and the resulting board state will be denoted by $R_n(S)$.

Lemma 2.4. *The stage one reduction is well defined. Namely, no q-stone is adjacent to components of both colors captured on this turn, so that the displayed conversion of q-stones is unambiguous; and the prunings of the entanglement state are unambiguous set operations.*

Proof. Suppose that a q-stone r is adjacent to unicolor components C_0, C_1 , both captured on this turn, with C_0 white and C_1 black. As C_1 is captured, it is in particular trapped. So C_0 is adjacent to the q-stone r , which is itself adjacent to the trapped component C_1 of the opposite color, and Condition 2 fails for C_0 . This contradicts the assumption that C_0 is captured.

Thus, each q-stone adjacent to some component captured on this turn is adjacent to captured components of only one color, and so its conversion is unambiguous. As the removal of stones is likewise unambiguous, each simultaneous resolution step, and hence the stage one reduction, is well defined. $\square$

If the stage one reduction removes no stones, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction, denoted by $R^k(S)$, satisfies:

$$R^1(S) = R_n(S).$$

$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

That is, at stage k the capture conditions of Definition 2.2 are applied with “this turn” read as turn $n-k+1$, on the current board state: a capture of an earlier turn, blocked at that turn, may complete at a later one, once the blocking entanglement

has collapsed. The recursion stops as soon as a stage removes no stones, and the turn is then resolved.

Remark 2.5. *In practice the resolution of most turns is easy to work out mentally; we will give some examples of “nested” entangled states further ahead.*

2.6. End of the game. An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some black/white stones on the board as prisoners for White/Black. (As is common in standard play of *Go*.) Each black or white prisoner gives one point to the player who holds it. This is the main way of ending the game.
- The display is unchanged after each of two consecutive turns. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row.
- The display has no empty intersections.

We then decide the winner using Chinese *Go*-style rules. So the winner is decided by the integer sum of the prisoner count and area count. We now explain the latter.

We say that an empty intersection i on the board **reaches black respectively white** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is **surrounded by black** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a q-stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

- (1) The number of black stones on the board.
- (2) The number of empty intersections surrounded by black.

See Section 3.5 for an example.

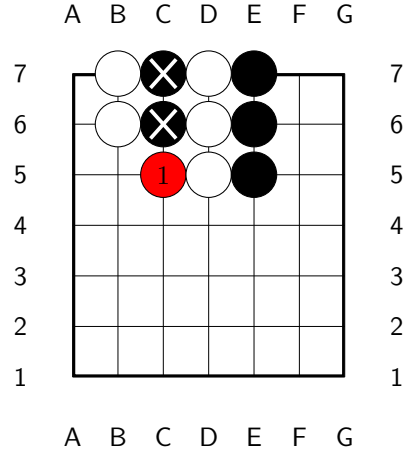
A draw is now possible, although compared to chess it is unlikely (for random games on the 19×19 board especially so, as the table in Section 2.1 shows).

3. EXAMPLES

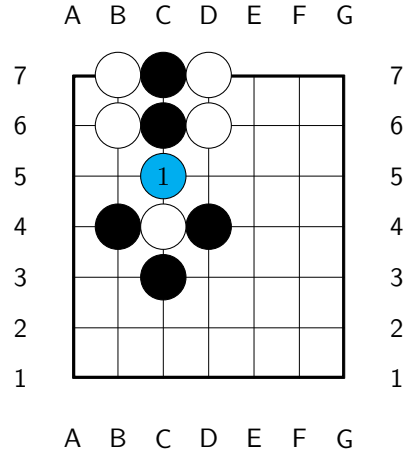
Throughout, a turn’s joint move is written m/m' , the move of the first player listed first – in *SGo*, Black’s.

3.1. Capture and non-capture moves. Here are some examples and non-examples of capture.

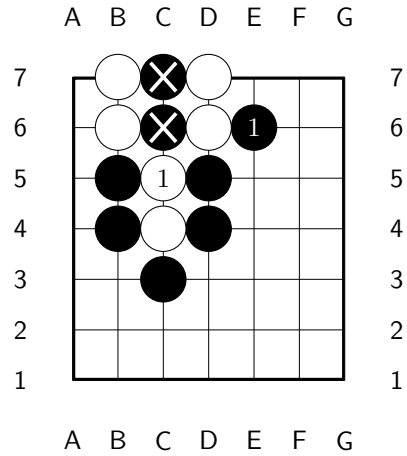
Turn 1 is $C5/C5$. This is a capture move, and C5 will resolve to white at the end of the turn.



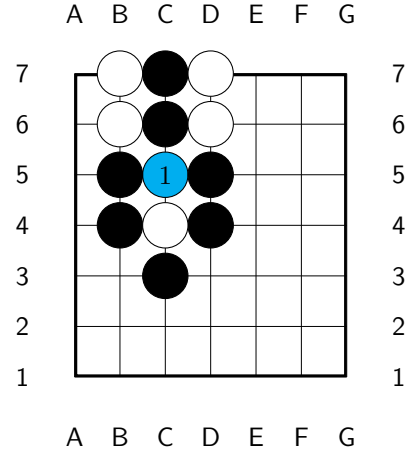
The turn is again $C5/C5$, in another position: not a capture move, as the created q-stone, marked blue, is adjacent to an opposite-colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



Turn 1 is $E6/C5$. White's move is a capture move, as Condition **3b** is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture.

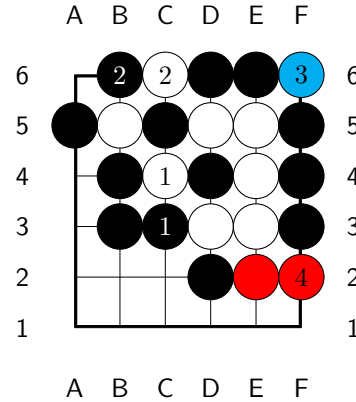


The turn is $C5/C5$: not a capture move, since Condition **3b** is now not satisfied, the stone placed on this turn being a q-stone (marked blue, as the blocker).

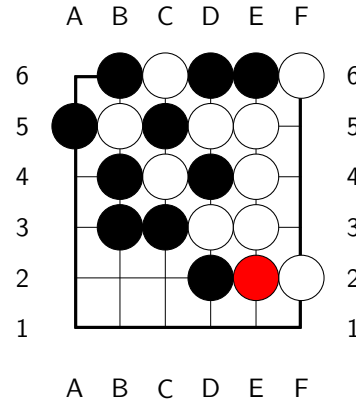


3.2. More complex examples.

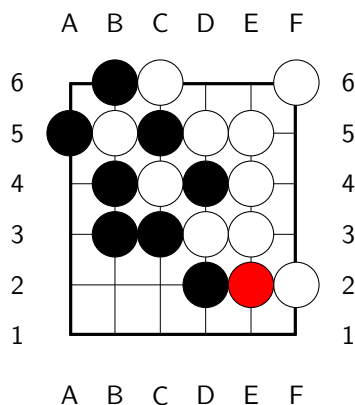
In the position on the right, turn 1 is $C3/C4$ – Black attempts capture at C3, White at C4. Nothing is captured on turn 1 according to our rules, and after turn 2 the position becomes an example of a “nested entangled state”. The continuation $B6/C6$; $F6/F6$; $F2/F2$ is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section; F6 is marked blue, being what blocks the capture of D6 and E6 on this turn. But F2 is a capture move for White since condition **3a** is satisfied.



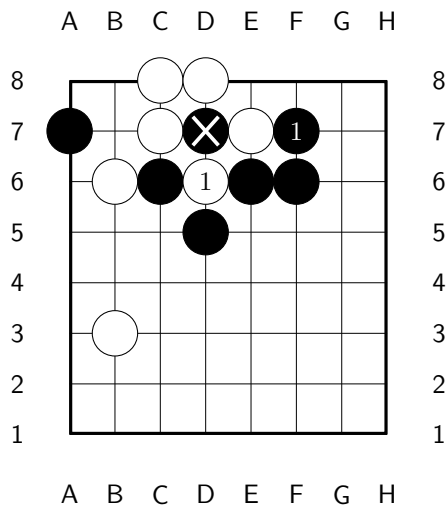
At the end of turn 4, stage one resolution is depicted on the right.



Stage two resolution. Stage three, which applies the capture conditions of turn 2, removes nothing: C5 and B5 each block the other's capture. So the recursion stops, and this is the final resolution of turn 4. Note that B5, C5, C4 and D4 are all trapped yet uncaptured – an entangled analogue of *seki*, impossible in classical *Go*: each capture is blocked by the entanglement of the others.

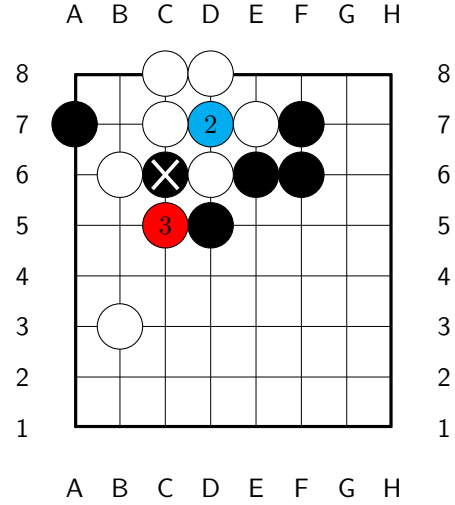


3.2.1. *Ko situation.* Below is a Ko type situation. Recall that we do not have a Ko rule; none is needed, since the board state cannot immediately loop by recapture.

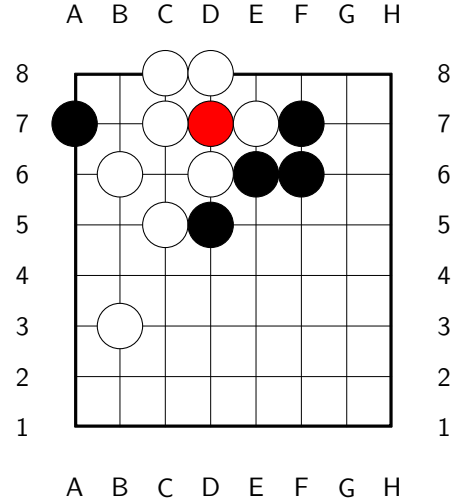


Turn 1 is $F7/D6$; White's D6 is a capture move.

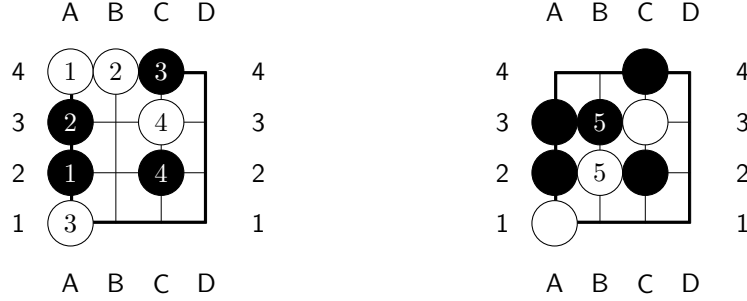
Here is the possible continuation $D7/D7$; $C5/C5$; the blue stone at D7 is what blocks Black's recapture of White's D6.



At the end of turn 3 we get the following position: the q-stone at D7 simply persists, and the Ko fight remains open, held in the entanglement.

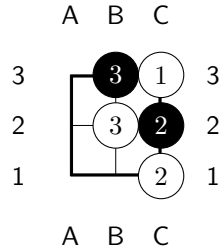


Finally, here is an example where the component clause of the suicidal capture condition **3b** – that the trapped neighbor's *component* contain a stone created on this turn – does the work. On the 4×4 board play the conflict-free history $A2/A4$; $A3/B4$; $C4/A1$; $C2/C3$:

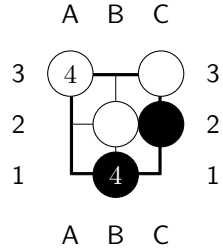


Now play $B3/B2$. Black's B3 fills the last liberty of the old white pair A4, B4, while White's B2 fills the last liberty of the black group A2, A3, B3. The black group is trapped by White on this turn, so condition 3a fails for the pair and the suicidal capture clause governs. The pair touches the trapped group at A3 – an old stone – but the group *contains* this turn's B3: the condition is met, and the pair is captured, as the right board shows. This is what the classical semantics demands: the two serializations of the turn agree, and each captures the pair – B3 played first captures it at once, and B2 then lands with the liberty B1; B2 played first, B3 still captures it, the black group surviving on the freed liberties. Had the clause read the adjacent stone alone, A3 being old, the capture would be refused, and the display would freeze on the board a pair that classical play removes in either order. The history is conflict-free, so the entanglement is a singleton throughout: the component clause is exactly what keeps the display classical here.

3.3. Quantum effects. The branches of the entanglement state can interact through the display; we might call these quantum effects. On the 3×3 board play the history $\text{pass}/C3; C2/C1; B3/B2$, arriving at the display:



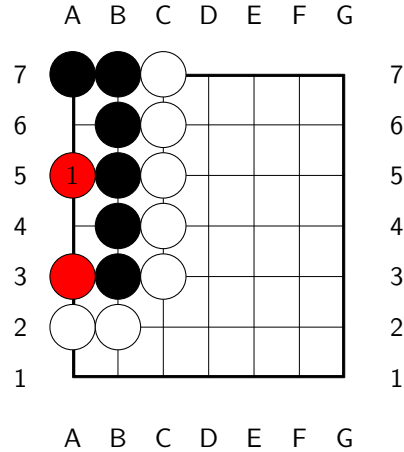
Both C3 and C2 are trapped yet uncaptured. In one reading of the entanglement Black's B3 has captured C3; in the other White's B2 has captured C2. This is a standoff on pure black and white stones, with no q-stone anywhere in the history. Now play the joint move $B1/A3$. The displayed resolution captures both B3 and C1:



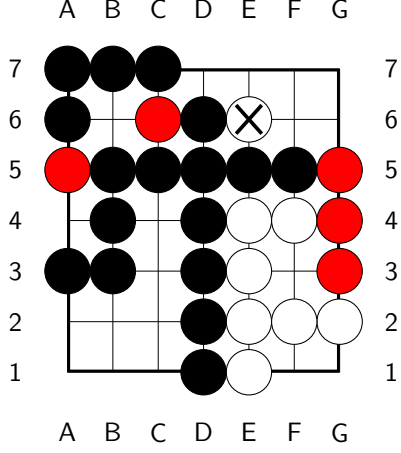
though each single reading captures only one of them. The other survives on a liberty freed by that reading's earlier capture. The display remembers both stones of the standoff. It thus performs a joint capture that no serialization of the history performs. Every reading is then incompatible with the display, and the entanglement reduces to the empty set. In the universal simultaneization of Section 4.4 the branches never interact: its evolution is the union of independent classical evolutions.

3.4. Life and death. Life and death is positionally very similar to *Go*; here is an example.

White cannot kill the black group: A4 and A6 are two eyes – a white stone placed in either is immediately captured, while a simultaneous conflict there yields a q-stone. In *SGo* the group is alive outright: even with Black's collaboration – Black filling one eye himself, White taking the other – the standing entanglement at A3 and A5 blocks every capture, and the group survives, wrapped in the entanglement.



3.5. End game position. In the diagram below, the players have agreed to end the game. The white stone at E6 is agreed to be a prisoner of Black. Assuming there are no other prisoners, White has $7+3$ points and Black has $7 \cdot 7 - (7+3) - 5 + 1$ points, where the -5 accounts for the five q-stones, which count for neither player, and the $+1$ for the prisoner.



4. SYMMETRIC SIMULTANEOUS COMBINATORIAL GAMES

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games. Such formalisms are standard in several communities: games with simultaneous moves appear as the deterministic case of the stochastic games of Shapley [21], and as the concurrent game structures of computer science [1]. Simultaneous play of combinatorial games as such is studied by Huggan–Nowakowski–Ottaway [9], see also the references therein; there, however, the effect of a pair of simultaneous moves must be specified by each ruleset separately, no general procedure being available. The emphasis here, on simultaneousizations of a given sequential game and on their simplicity, appears to be new, and in this setting the pseudo pass moves together with the operation $\#$ further below do furnish such a general procedure. The inverse passage, representing a sequential game within an intrinsically simultaneous formalism, is standard in general game playing, and is likewise effected by pass-like moves: there the idle player is required to play an explicit “noop” move, cf. [22].

Definition 4.1. A *combinatorial simultaneous game* (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the *pseudo pass move* (it has multi grading both 0 and 1).
- Game states decompose as

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where $B(G)$ is some set.

- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.

- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
- (3) If s is not final, $E(s, 0)$ is defined.
- (4) If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' is 0).
- (5) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined, then $E(s, m_0, m_1)$ is defined.
- (6) For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

We say that G is **finite** if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \ E(q_0, m_1, \dots, m_k) \text{ is defined.}$$

Note that any game history of length $2k$ can be rewritten in the form $h = \{m_i^0, m_i^1\}_{i=1}^{i=k}$, with $m_i^j \in M_j(G)$.

It is our convention that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(q_0, m_1, \dots, m_k)$, for some m_i .

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4.2. We say that a combinatorial simultaneous game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{A} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
- (2) There is a bijection $\mathcal{R} : B(G) \rightarrow B(G)$, satisfying $\mathcal{R}^2 = id$ and $\mathcal{R}(q_0) = q_0$, which we extend to $\mathcal{R} : S(G) \rightarrow S(G)$ by $\mathcal{R}(s, m) = (\mathcal{R}(s), \mathcal{A}(m))$, s.t. for all $s \in S(G)$, all $m \in M(G)$, and for $\mathcal{A}$ as above we have:
 - $\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
 - $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

4.1. Formalizing SGo . A state s of SGo is a pair. The first member is the display of Section 2, together with the time stamps of its stones. The second is the **entanglement state** $\rho(s)$: a finite set of classical Go positions, called **branches** – the classical positions that the play so far may have produced. The initial entanglement state is the singleton of the empty position. On a turn, first each branch

evolves as a classical *Go* position: the two moves of the turn are played on it in both orders, a move to an occupied intersection forfeited, and the outcomes are collected. The display is resolved by the rules of Section 2, and the resolved display prunes: the new entanglement state consists of the collected branches **compatible** with it. Compatible means: every stone of the branch stands on an occupied intersection of the display, and matches the color of the displayed stone there, a q-stone matching either color. As with chess, whose position includes the castling and en passant rights in addition to the piece diagram, the entanglement state is information the diagram does not show; it can be recovered from the recorded game history.

Lemma 4.3. *SGo is an ssc game, that is a symmetric simultaneous combinatorial game.*

Proof. Set $G = SGo$. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pseudo pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is the full data maintained by the rules of Section 2: the display with its placement records, together with the entanglement state of Section 4.1 – a finite set of legally obtainable positions of *Go*. The initial state is the empty display with the entanglement $\{q_0\}$, for q_0 the empty position.

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are moves to empty intersections of the display, then $E((s, m_0), m_1) \in B(G)$ is defined to be the state produced by the resolution rules of Section 2: the display resolves by the placement and capture rules. Each branch evolves under the serialized moves, and the branches incompatible with the resolved display are discarded.

To see that G is symmetric, let $\mathcal{R} : B(G) \rightarrow B(G)$ be the map that interchanges the black and white colors of the stones in every branch (on the display this interchanges black and white and leaves q-stones invariant). And let $\mathcal{A} : M(G) \rightarrow M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \rightarrow M_1(G)$, which then induces $\mathcal{A}$. Then clearly the conditions of Definition 4.2 will be satisfied. $\square$

Remark 4.4. *We have not formalized $F(SGo)$: ending a game by agreement is not an automaton notion. This caveat is however minor, and is already present in classical *Go*.*

4.2. Solvability. A theorem of Zermelo [20] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly *SGo* cannot have a winning strategy: mirroring a winning strategy via the symmetry operators would yield one for the opponent, and both players cannot win. Furthermore, unless we put an upper bound on the length of a game, or on the number of position repetitions, an *SGo* game can be infinitely long.⁶ So we will assume there is such a bound, which will make *SGo* a finite game.

One may ask whether *SGo* has a *strategy to draw*: a pure strategy guaranteeing at least a draw. Exact computation shows that on the 2×2 board it does – both

⁶This is also true for *Go* and chess.

players can force a draw with any game length bound between 6 and 10 – but already on the 3×3 board, with any bound between 6 and 8, neither player has such a strategy: a player following any fixed pure strategy can be made to lose. Optimal play is thus necessarily mixed from the 3×3 board on, and we expect the same on all larger boards – the mixed equilibrium of Theorem 4.5 below is not a formality.

Theorem 4.5. *For any assignment of payoffs to a win, a loss and a draw, a finite combinatorial simultaneous game G (Definition 4.1) has a Nash equilibrium.*

Proof. For a non final $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \mid E(s, m) \text{ is defined}\},$$

which is non-empty as it contains the pseudo pass move 0.

A **pure strategy** of player P_i assigns to each non final $s \in B(G)$ a move in $M_i(s)$, so that the set of pure strategies is the finite set

$$\Sigma_i = \prod_s M_i(s),$$

the product over the non final elements of $B(G)$. A pair of pure strategies determines a play of the game, and so a payoff, by finiteness of G .⁷ This presents G in normal form: a finite two-player game with the strategy sets Σ_0, Σ_1 . The mixed strategies of player P_i are then $\mathcal{P}_i = \Delta(\Sigma_i)$, the probability distributions over Σ_i . The theorem is now a direct application of the Nash existence theorem [13]. $\square$

Corollary 4.6. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 for a draw. By the normal form presentation in the proof of Theorem 4.5, G is a zero sum matrix payoff game, with the rows and columns of the payoff matrix indexed by the pure strategies Σ_0, Σ_1 . By the classical theory of matrix payoff games (see for instance [14]), all Nash equilibria have the same expected payoff for Black, namely the value v of the matrix game, and an equilibrium strategy of Black guarantees Black an expected payoff of at least v against every strategy of White. So it is enough to show that $v = 0$; that a symmetric zero sum game has value zero is classical, cf. [7], and for completeness we give the argument.

Suppose otherwise, and suppose without loss of generality that $v > 0$, so that for an equilibrium joint strategy

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

Black has expected payoff $v > 0$.

Let $\mathcal{R}, \mathcal{A}$ be the symmetry operators as in the proof of Lemma 4.3. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has expected payoff v , that is with Black expected payoff $-v < 0$. As the latter must again be v , we have a contradiction. $\square$

⁷Possibly the finiteness condition can be relaxed.

4.3. Symmetric combinatorial sequential games. Combinatorial sequential games have a beautiful Conway recursive definition [4]. However, for our purpose, in particular for the universal simultaneization construction, it is convenient to interpret them in terms of finite automata.

Definition 4.7. A *combinatorial sequential game* (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the *pseudo pass move* (it has multi grading both 0 and 1).
- A $\mathbb{Z}_2$ graded set of game states $S(G) = \text{States}(G) = S_0(G) \sqcup S_1(G)$, we let $\text{deg} : S(G) \rightarrow \mathbb{Z}_2$ denote the associated function.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S_0(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) $E(s, m)$ is defined only if m has the same grading as s or if $m = 0$.
- (3) If $E(s, m)$ is defined then $\text{deg}(E(s, m)) = \text{deg}(s) + 1$.
- (4) If s is not a final state then $E(s, 0)$ is defined.
- (5) If $E(s, 0) = E(s', 0)$ then $s = s'$.
- (6) If $E(s, 0, 0)$ is defined then it is a final state: two consecutive pseudo pass moves end the game.

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

As before it will be understood that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(s, m_1, \dots, m_k)$, for some m_i .

Example 4.8. Note that Go has a pseudo pass move, but chess does not; however, chess can be augmented with a pseudo pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules, and with the game ending in a tie after two consecutive passes. This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in *Zugzwang* they may just pass). We will call this **augmented chess** denoted by *chess*.

Definition 4.9. We say that a combinatorial sequential game G is *symmetric* if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{A} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{A}^2 = \text{id}$, $\mathcal{A}(0) = 0$.

- (2) There is a bijection $\mathcal{R} : S(G) \rightarrow S(G)$, satisfying $\mathcal{R}^2 = id$ and taking $S_0(G)$ to $S_1(G)$ s.t. for all $s \in S(G)$, all $m \in M(G)$ and for $\mathcal{A}$ as above we have:

$$\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m),$$

whenever both sides are defined. Furthermore, if one side is defined then so is the other.

- (3) $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Example 4.10. *Of course $\widetilde{\text{chess}}$ and classical Go are examples of symmetric games. In the case of Go , $\mathcal{A}$ interchanges black and white moves and $\mathcal{R}$ likewise interchanges the color of the stones on the board and shifts the initiative (it is easy to see that this does result in a legally reachable position).*

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{A}$, $\mathcal{R}$ implicitly fixed).

4.4. Universal simultaneization. Let G be a sequential game (by our standing assumption with a pass move 0 and symmetric). Let $\{m_i\}$ be a sequence of game moves. If some $m_i = 0$ and $m_k \neq 0$ for some $k > i$, then we say that it is an **interior pass move**. We say that a finite sequence $\{m_i\}$ of moves of G is **pseudo legal** if it becomes a game history after adding or removing some interior pass moves from the sequence.

For $a \in S(G)$ if there exists an $a' \in S(G)$ s.t. $E_G(a', 0) = a$ we set $\bar{a} = a'$, otherwise set $\bar{a} = E_G(a, 0)$. Property 5 in Definition 4.7 ensures that it does not matter which a' is chosen when it exists. For $a \in S(G)$ and $m \in M(G)$ set

$$a \# m = \begin{cases} E_G(a, m), & \text{if this is defined} \\ E_G(\bar{a}, m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Let $\sim$ denote the **pass equivalence**: the equivalence relation on the states of G generated by $s \sim E(s, 0)$, for s and $E(s, 0)$ non final. We call its classes **pass classes**. By Property 5 and Property 6, a pass class contains at most one state of each grading. For a non final $b \in S(G)$ we let $\hat{b}$, the **grading normalization** of b , denote the grading 0 member of the pass class of b when it exists, and $\hat{b} = E_G(b, 0)$ otherwise; so $\hat{b} = b$ for $b \in S_0(G)$, and $\hat{b}$ is pass equivalent to b outside of the degenerate second case, in which a pass is appended. It is convenient to regard final states as carrying both gradings, as with the pseudo pass move, and for final b we set $\hat{b} = b$. Define $P(a, m_0, m_1) \in 2^{S_0(G)}$, for $a \in S_0(G)$, by the following prescription in decreasing priority, where each term is understood to be replaced by its grading normalization, and the expression is defined if and only if all the terms are defined.

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1 \# m_0$ (note that if $m_0 = m_1 = 0$ then the two terms coincide, so that $P(a, 0, 0) = a \# 0 \# 0$.)
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S_0(G) \times M_0(G) \times M_1(G) \rightarrow 2^{S_0(G)},$$

which we call the **serialization map**: it evaluates the turn in its available serializations – both orders of play when both are defined, the degenerate cases otherwise – and collects the outcomes.

Definition 4.11. Let G be a sequential game. Its **universal simultaneization** is the simultaneous game $\text{Sim}(G) = G'$ satisfying the following.

- (1) $B(G') = 2^{S_0(G)}$. (The elements s of $\mathfrak{s} \in B(G')$ will be called **state branches**.) By the above, an element of $B(G')$ may equivalently be regarded as a finite collection of pass classes, represented by their grading 0 members.
- (2) The initial state $q'_0 = \{q_0\}$.
- (3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
- (4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$; then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$, s is not final and $s \# m_i$ is defined.
- (5) For $\mathfrak{s}$ and m_i as just above:

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} P(s, m_0, m_1),$$

(undefined if some $P(s, m_0, m_1)$ is undefined).

(6)

$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s} \, s \in F(G).$$

Note that $\text{Sim}(G)$ is uniquely defined by the above conditions up to the partition of its final states into W_0, W_1, D . The partition plays no role beyond the symmetry of Lemma 4.13; one may take $\mathfrak{s}$ winning for a player exactly when some branch wins for them and no branch wins for the opponent, and drawn otherwise.

Remark 4.12. *The passage from positions to sets of positions is a form of the classical subset construction of automata theory [16]. In game theoretic terms, a state of $\text{Sim}(G)$ is a knowledge state: the set of classical positions consistent with what the players have observed. Analogous knowledge based constructions are used in the study of games of imperfect information, see for instance [18], [5].*

Lemma 4.13. $\text{Sim}(G)$ is symmetric.

Proof. Set $G' = \text{Sim}(G)$, and let $\mathcal{A}, \mathcal{R}$ be the symmetry operators of G . Set $\tilde{\mathcal{A}} : M(G') \rightarrow M(G')$ to just be $\mathcal{A}$. For

$$\{s_0, \dots, s_k\} = \mathfrak{s} \in B(G'),$$

set

$$\tilde{\mathcal{R}}(\mathfrak{s}) = \{\mathcal{R}(s_0), \dots, \mathcal{R}(s_k)\},$$

with each $\mathcal{R}(s_i)$ replaced by its grading normalization, $\mathcal{R}$ being grading reversing. Since $\mathcal{A}(0) = 0$, the map $\mathcal{R}$ commutes with the pass move and so preserves pass equivalence; with Property 6 it follows that $\tilde{\mathcal{R}}^2 = id$ exactly, and that the computation below is unaffected by the normalization. For

$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$

set

$$\tilde{\mathcal{R}}(\mathfrak{s}, m) = (\tilde{\mathcal{R}}(\mathfrak{s}), \tilde{\mathcal{A}}(m)).$$

The needed symmetry properties for the maps $\tilde{\mathcal{A}}, \tilde{\mathcal{R}}$ are then verified using the symmetry properties of $\mathcal{A}, \mathcal{R}$ as follows. It is easily seen that $\overline{\mathcal{R}(a)} = \mathcal{R}(\bar{a})$, hence

$$\mathcal{R}(a) \# \mathcal{A}(m) = \begin{cases} E_G(\mathcal{R}(a), \mathcal{A}(m)) = \mathcal{R} \circ E_G(a, m), & \text{if defined} \\ E_G(\mathcal{R}(\bar{a}), \mathcal{A}(m)) = \mathcal{R} \circ E_G(\bar{a}, m), & \text{if defined} \end{cases}$$

and so

$$(4.14) \quad \mathcal{R}(a) \# \mathcal{A}(m) = \mathcal{R}(a \# m),$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$; then we get:

$$\begin{aligned} \tilde{\mathcal{R}}(E_{G'}(\tilde{\mathcal{R}}(\mathfrak{s}), \tilde{\mathcal{A}}(m_0), \tilde{\mathcal{A}}(m_1))) &= \tilde{\mathcal{R}} \circ \cup_{s \in \mathfrak{s}} P(\mathcal{R}(s), \mathcal{A}(m_0), \mathcal{A}(m_1)) \\ &= \tilde{\mathcal{R}} \circ \cup_{s \in \mathfrak{s}} \mathcal{R} \circ P(s, m_0, m_1) \text{ using (4.14)} \\ &= \cup_{s \in \mathfrak{s}} P(s, m_0, m_1) \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies property 2 of Definition 4.2 for pair moves at states of $B(G')$; the rest is similar. $\square$

Definition 4.15. Let G_0 be a sequential game and G_1 a symmetric simultaneous game with $M(G_1) = M(G_0)$. The **semi-classical** states of G_1 , with their **associated positions** in G_0 , are defined recursively: the initial state of G_1 is semi-classical, with associated position the initial state of G_0 . If s is semi-classical with associated position a , and m_0, m_1 commute at a – both $a \# m_0 \# m_1$ and $a \# m_1 \# m_0$ are defined and pass equivalent – then $E_{G_1}(s, m_0, m_1)$, if defined, is semi-classical, with associated position the common grading normalization of $a \# m_0 \# m_1$ and $a \# m_1 \# m_0$.

Definition 4.16. Let G_0 be a sequential game. We say that G_1 is a **simultaneization** of G_0 if the following is satisfied.

- (1) G_1 is a symmetric simultaneous game.
- (2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (3) There is a map $\rho : B(G_1) \rightarrow B(\text{Sim}(G_0))$ satisfying $\rho(q_0) = q'_0$ and s.t. for $s \in B(G_1)$: whenever the state $E_{G_1}(s, m_0, m_1)$ is defined, so is $E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1)$, and

$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1).$$

- (4) For every semi-classical s with associated position a (Definition 4.15): $\rho(s) = \{a\}$.

For $a \in S(G_0)$ set

$$\mathbb{M}(a) = \{m \in M(G_0) \mid a \# m \text{ is defined}\},$$

which we call the **interface** of the position a : the set of moves of either player available at a , with the operation $\#$ absorbing the turn order. Likewise, for $s \in B(G_1)$ set $M(s) = \{m \in M(G_1) \mid E_{G_1}(s, m) \text{ is defined}\}$, the interface of the state s .

Lemma 4.17. *Let G_1 be a simultaneization of G_0 . Then for every non final $s \in B(G_1)$: every branch of $\rho(s)$ is non final, and*

$$M(s) \subseteq \bigcap_{a \in \rho(s)} \mathbb{M}(a).$$

That is, a move available at s is available in every branch. Equality of the two sides, together with $\rho(s) \neq \emptyset$, is precisely the statement that a joint move is available at s if and only if it is available in $\text{Sim}(G_0)$ at $\rho(s)$.

Proof. First, for $m \in M(G_1)$, $E_{G_1}(s, m)$ is defined if and only if the pair move $E_{G_1}(s, m, 0)$ (in the order of the pair prescribed by the grading of m) is defined. Indeed, the latter presupposes the former, and conversely if $E_{G_1}(s, m)$ is defined then so is $E_{G_1}(s, 0)$, as s is not final, and so by the properties of Definition 4.1 the pair move is defined, using that the pseudo pass move has both gradings. If some $a \in \rho(s)$ were final, then no joint move would be defined in $\text{Sim}(G_0)$ at $\rho(s)$, by the construction of $\text{Sim}(G_0)$; by condition 3 no joint move would then be defined at s , contradicting the definedness of the pair $(0, 0)$ at the non final s . So every branch is non final. Now if $m \in M(s)$, the pair move is defined at s , hence, by condition 3, defined in $\text{Sim}(G_0)$ at $\rho(s)$, which says exactly that $a \# m$ and $a \# 0$ are defined for every $a \in \rho(s)$, that is $m \in \mathbb{M}(a)$ for every a . The equality statement is the same unwinding read in both directions. $\square$

Definition 4.18. Let G_1 be a simultaneization of G_0 . We say that G_1 is **faithful** at a state $s \in B(G_1)$ if $\rho(s)$ is nonempty, s is final only if some branch of $\rho(s)$ is final, and, when s is not final,

$$M(s) = \bigcap_{a \in \rho(s)} \mathbb{M}(a),$$

that is, exactly the joint moves of $\text{Sim}(G_0)$ at $\rho(s)$ are available at s . The finality clause says that G_1 may not invent endings where it is faithful. For $d \geq 0$, we say that G_1 is **faithful within distance d of classical play** if it is faithful at every state reachable from a semi-classical state in at most d turns, distance 0 being the semi-classical states themselves. The **faithfulness radius** of G_1 is the supremum of the d such that G_1 is faithful within distance d of classical play, possibly 0 or ∞ : the depth to which its branch reduction stays exact. It is defined when G_1 is faithful within distance 0; otherwise G_1 has no radius.

Definition 4.19. Let G_0 be a sequential game. We will say that a simultaneization G_1 of G_0 is a **simple simultaneization of G_0** if for every non final $s \in B(G_1)$ at which a non-pass move is available there is a legally obtainable position $a \in S(G_0)$ with

$$\mathbb{M}(a) = M(s),$$

where for s the initial state of G_1 , a can be taken to be q_0 . The **absolute radius** of G_0 is the supremum of the faithfulness radii of its simple simultaneizations; a simultaneization with no radius contributes nothing.

Remark 4.20. *Both radii are isomorphism invariants. An isomorphism of games – a bijection of states and of graded moves, intertwining the evolutions, the pass moves, and the final states – preserves commutation, semi-classicality, interfaces, and simplicity. So compatibly isomorphic simultaneizations have equal faithfulness*

radii, and isomorphic sequential games have equal absolute radii: the absolute radius measures the game, not its presentation.

5. MAIN THEOREMS

5.1. The decoherent variant $DSGo$. We now introduce a theoretical variant $DSGo$ ⁸: SGo with the capture resolution replaced by decoherence. Each turn, the stones are placed by the placement rule of SGo , and nothing else is played out on the display. The resulting state is then postprocessed as follows. The display is measured against the branch evolution of the turn; the entanglement does the measuring and the removing. A state of $DSGo$ carries the same data as one of SGo – a display together with its entanglement state $\rho(s)$ – and moves are available as in SGo .

The serialization map extends to sets of positions elementwise,

$$P(\mathfrak{B}, m_0, m_1) = \bigcup_{a \in \mathfrak{B}} P(a, m_0, m_1);$$

we call $P(\rho(s), m_0, m_1)$ the *decoherence* of the turn m_0/m_1 at s . Next, the *postprocessing* $\Delta(s', \mathfrak{M})$ of a state s' by a finite set $\mathfrak{M}$ of positions is defined as follows. On the display of s' , a stone whose intersection is occupied in no element of $\mathfrak{M}$ is removed, and a q-stone created on this turn carrying a single color over the elements of $\mathfrak{M}$ becomes that color; the entanglement state is then re-cut to the elements of $\mathfrak{M}$ occupying no empty intersection of the display. This round is repeated, with the re-cut entanglement in place of $\mathfrak{M}$, until it removes no stone; as with the stage recursion of Section 2.5, the repetition is finite. The evolution of $DSGo$ is the postprocessed placement:

$$E_{DSGo}(s, m_0, m_1) = \Delta(\text{Pl}(s, m_0, m_1), P(\rho(s), m_0, m_1)),$$

where $\text{Pl}(s, m_0, m_1)$ is s with the turn's stones placed by the placement rule of SGo and no capture resolution applied.

As the proof of Theorem 5.2 shows, the postprocessing loses no element of the decoherence. Hence $\rho \circ E_{DSGo} = E_{\text{Sim}(Go)} \circ \rho$: the entanglement component of a $DSGo$ history is a $\text{Sim}(Go)$ history. $DSGo$ referees $\text{Sim}(Go)$ on a display, differing from it only in scoring.

The construction interpolates. For $k \geq 1$, the delayed game k - $DSGo$ withholds the objective reduction for k turns: a state carries, besides the display and $\rho(s)$, the stored resolutions of the last k turns. On a turn, the stored resolution of the turn played k turns earlier is executed first: its removals and conversions are applied to the stones that still stand, a stone being identified by its intersection and its time stamp. The turn's stones are then placed, and the display is postprocessed against the decoherence of the turn, as in $DSGo$. Finally, the capture resolution of the resulting display – the stage recursion of Section 2.5, anchored at this turn – is computed and stored, and nothing of it is played now. As $k \rightarrow \infty$ no stored resolution ever executes, and the game is $DSGo$; SGo , which resolves at once and carries no decoherence, stands at the origin of the ladder.

For an intersection i and a classical position b , let $ev_i(b) \subseteq \{\text{black}, \text{white}\}$ be the color of the stone of b at i , the empty set if i is empty in b . Extend ev_i to a

⁸ D stands for decoherence. Not meant to be humanly playable without computer resolution.

set of positions by union, and to a state by reading its display, a q-stone reading {black, white}.

Lemma 5.1. *Let s be a state and $\mathfrak{M}$ a finite set of positions.*

- (1) *If $b_0 \in \mathfrak{M}$ satisfies, for every intersection i , $ev_i(s) = \emptyset \implies ev_i(b_0) = \emptyset$, then $b_0 \in \rho(\Delta(s, \mathfrak{M}))$.*
- (2) *For every intersection i : $ev_i(\Delta(s, \mathfrak{M})) = \emptyset \iff ev_i(\rho(\Delta(s, \mathfrak{M}))) = \emptyset$.*

Proof. Let $D_k, \mathfrak{M}_k$ be the display and the entanglement after k rounds, with D_0 the display of s and $\mathfrak{M}_0 = \mathfrak{M}$. A round removes the stones at the intersections i with $ev_i(\mathfrak{M}_k) = \emptyset$, and re-cuts:

$$\mathfrak{M}_{k+1} = \{b \in \mathfrak{M}_k \mid \forall i : ev_i(D_{k+1}) = \emptyset \implies ev_i(b) = \emptyset\}.$$

(The recoloring clause does not change occupancy and plays no role.) For (1) we show by induction on k : $b_0 \in \mathfrak{M}_k$, and $ev_i(D_k) = \emptyset \implies ev_i(b_0) = \emptyset$ for all i . For $k = 0$ this is the hypothesis. For the step: an intersection emptied by round $k + 1$ has $ev_i(\mathfrak{M}_k) = \emptyset$, and $ev_i(b_0) \subseteq ev_i(\mathfrak{M}_k)$ since $b_0 \in \mathfrak{M}_k$. With the hypothesis, $ev_i(D_{k+1}) = \emptyset \implies ev_i(b_0) = \emptyset$, hence $b_0 \in \mathfrak{M}_{k+1}$.

For (2): let D and $\mathfrak{M}_*$ be the final display and entanglement, so $\mathfrak{M}_* = \rho(\Delta(s, \mathfrak{M}))$. If $ev_i(D) = \emptyset$ then $ev_i(b) = \emptyset$ for every $b \in \mathfrak{M}_*$, by the re-cut; so $ev_i(\mathfrak{M}_*) = \emptyset$. Conversely, a stone of D at an intersection i with $ev_i(\mathfrak{M}_*) = \emptyset$ would be removed by a further round, and the repetition has stabilized. $\square$

Let Go be the classical game with Chinese style area scoring and standard pass move, with the following stipulations: suicide is a legal move, as for example in the Tromp–Taylor rules [24, 23], there is no Ko or superko rule, and the game is ended analogously to the way SGo is ended, so that the role of a superko rule is played by the end of the game rules.

Theorem 5.2. *SGo , the delayed games k - $DSGo$ for $k \geq 1$, $DSGo$, and $\text{Sim}(Go)$ are simple simultaneizations of Go . Their faithfulness radii are, respectively, zero, exactly k , infinity, and infinity.*

Proof. Conditions 1 and 2, for SGo , $DSGo$ and the k - $DSGo$, are by construction and a direct analogue of Lemma 4.3.

Let $\rho : B(SGo) \rightarrow B(\text{Sim}(Go))$ send a state to its entanglement component: the entanglement state is a finite set of positions of Go , that is an element of $B(\text{Sim}(Go))$ (positions of Go being identified with their pass classes, via the grading normalization in the definition of P). Then $\rho(q_0) = \{q_0\} = q'_0$. The branch evolution of Section 4.1 is precisely the evolution by the serialization map P : playing the two moves in both orders, with a move to an occupied intersection forfeited, is the prescription defining P . So $E_{SGo}(s, m_0, m_1)$ is obtained by applying the resolution prunings to $E_{\text{Sim}(Go)}(\rho(s), m_0, m_1)$, and the inclusion in condition 3 holds by construction.

For the definedness direction of condition 3: a move of SGo is available if and only if its intersection is empty on the display; a branch kept by the reduction has all its stones within the display, a branch occupying a display-empty intersection being incompatible with the display and discarded; so a display-empty intersection is empty in every branch, and the joint move is available in $\text{Sim}(Go)$ at $\rho(s)$.

For the semi-classicality condition: along a generating history of Definition 4.15 every turn commutes, so each branch evolution produces a single position $\hat{a}$, and

the entanglement state remains a singleton. For *DSGo* this is a proof: inductively the display is the diagram of the associated position, the placements match, and the postprocessing against $\{\hat{a}\}$ removes exactly the stones $\hat{a}$ lacks and keeps $\hat{a}$ in the re-cut, by Lemma 5.1(1); so the display is the diagram of $\hat{a}$ and $\rho(s)$ is the required singleton.

For *k-DSGo* one line is added: the display of a commuting turn is the diagram of a legal position, every component of which has a liberty, so the stage recursion captures nothing and the stored resolution is empty; the executions change nothing, and the proof above applies.

For *SGo*: at these turns the two serializations agree, and the capture resolution is designed to perform exactly the captures of this common serialization: a stone is removed only if it is removed there, and every capture there is performed, the captured group having lost its last liberty to a stone placed on the turn.⁹ A complete formalization of this claim would amount to re-deriving the capture conditions of Section 2 in the automaton formalism, which we do not undertake. On the 3×3 board it holds exhaustively: as computer verified at every legal classical position, every commuting pair resolves the display to the common outcome. Granting it, the display of *SGo* is, inductively, the diagram of $\hat{a}$, and the pruning keeps $\hat{a}$: $\rho(s) = \{\hat{a}\}$.

To verify the simplicity property: since in our version of *Go* suicide is a legal move and there is no Ko rule, for any non final position a of *Go* the interface $\mathbb{M}(a)$ consists of the pass moves and the moves of either player to the empty intersections of a . Likewise, for a non final $s \in B(\textit{SGo})$, $M(s)$ consists of the pass moves and the moves of either player to the empty intersections of s . So it suffices to exhibit a legally obtainable position $\phi(s)$ of *Go* with the same empty intersections as s . Let $\phi(s)$ be obtained by replacing every stone of s by a black stone. To see that $\phi(s)$ is legally obtainable, fix an empty intersection e of s , which exists as a non-pass move is available at s , and let Black play the stones of $\phi(s)$ in order of weakly decreasing distance to e , while White always passes. When a stone is placed, a geodesic from it to e exits the set of already placed stones at an empty intersection adjacent to the set, so at every moment every black component has a liberty, and no captures or suicides occur.

The simplicity argument applies verbatim to *DSGo* and the *k-DSGo*. For these games, the inclusion of condition 3 is the re-cut of Δ , by definition; and a surviving element occupies only display-occupied intersections, which gives the definedness direction as before.

For $\text{Sim}(\textit{Go})$, take ρ the identity. It is a symmetric simultaneous game with the moves of *Go*: Definition 4.11 and Lemma 4.13. Condition 3 is an equality. The semi-classicality condition follows from the recursion of Definition 4.15: a commuting pair evolves a singleton to a singleton. By construction, a move is available at a state $\mathfrak{s}$ if and only if it is available in every branch: $M(\mathfrak{s}) = \bigcap_{b \in \mathfrak{s}} \mathbb{M}(b)$. Also by construction, $\mathfrak{s}$ is final if and only if some branch is final. So $\text{Sim}(\textit{Go})$ is faithful at every reachable state, and its faithfulness radius is infinite. For its simplicity:

⁹In the first example of Section 3.1 the resolution performs the captures of one serialization and discards the other reading: “every” fails. In a mutual capture race, where each serialization captures a side, the resolution captures neither: the converse fails, by design. At entangled states the single witness itself fails, and the example of Section 3.3 is built on its failure: each of its two captures is performed by some reading, but no single reading performs both, and the entanglement empties.

when a non-pass move is available at $\mathfrak{s}$, some intersection is empty in every branch. The monochrome position with a black stone on every other intersection has the interface $M(\mathfrak{s})$, and it is legally obtainable by the geodesic construction above. So $\text{Sim}(Go)$ is simple.

It remains to compute the radii of SGo , $DSGo$ and the k - $DSGo$.

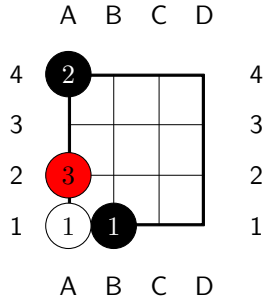
For SGo : at a semi-classical state s with associated position a , the display is the diagram of a and $\rho(s) = \{a\}$, as established above; so $M(s) = \mathbb{M}(a)$, and the finality clause holds, the end of the game rules of the two games being matched by the convention fixed above. Thus SGo is faithful within distance zero. It is not faithful within distance one, on any board: reach, by commuting turns, black stones on A3, B2, C1 and white stones on A2, B1, and play the conflict A1/A1. Black's reading captures both white stones; White's reading is a suicide; and the created q-stone is adjacent to an opposite-colored trapped component, so the resolution removes nothing (compare the second example of Section 3.1). So A2 and B1 stand on the display yet are empty in every element of the entanglement: $M(s') \subsetneq \bigcap_{b \in \rho(s')} \mathbb{M}(b)$. The radius of SGo is zero.

For the radius of $DSGo$: we first formalize final states of $DSGo$ as follows. A state s of $DSGo$ is final exactly when some branch of $\rho(s)$ is final (of course this requires fixing a formalization of final states of Go). Then the finality clause holds by definition.

For the move clause, at any reachable state s and any turn: by the re-cut of Δ , every element of $\rho(s)$ occupies only display-occupied intersections. The placement empties no intersection, so every element of the decoherence $P(\rho(s), m_0, m_1)$ has its stones among the display stones and the placements. Each element therefore satisfies the hypothesis of Lemma 5.1(1), and survives: $\rho(s')$ is the entire decoherence, and it is nonempty. By Lemma 5.1(2), an intersection is empty on the display of s' if and only if it is empty in every element of $\rho(s')$, that is $M(s') = \bigcap_{b \in \rho(s')} \mathbb{M}(b)$: faithfulness at s' . So $DSGo$ is faithful at every state reachable in play, and its faithfulness radius is infinite.

For the radius of k - $DSGo$: within distance k of classical play, every resolution that executes was stored at a turn of the generating commuting history, and is empty, as shown above. So every turn within this distance is a turn of $DSGo$, and the preceding argument applies: k - $DSGo$ is faithful within distance k .

It is not faithful within distance $k + 1$, on any board. Reach, by commuting turns, a black stone on B1 and a white stone on A1 – with a black stone on A4 on boards beyond 3×3 – and play the conflict A2/A2:



8	.	n	b	q	k	b	n	r
7	.	p	p	p	p	p	p	p
6	.	r	.	.	.	.	.	.
5	p	.	.	.	.	.	.	.
4	P	.	.	.	.	.	.	.
3	.	R	.	.	.	.	.	.
2	.	P	P	P	P	P	P	P
1	.	N	B	Q	K	B	N	R
	a	b	c	d	e	f	g	h

FIGURE 1. The position a^* , reached by the mirror pairs $a4/a5$; $Ra3/Ra6$; $Rb3/Rb6$. White pieces are in capitals. The conflicting simultaneous moves are the rook moves $m_0 : b3 \rightarrow b5$ and $m_1 = \mathcal{A}(m_0) : b6 \rightarrow b4$.

Its stored resolution captures A1 and converts the q-stone black: the black reading of the conflict performs exactly this capture, while the white reading joins A1 and lives. After $k-1$ turns of double passes, play B3/A3. The stored resolution executes, at distance $k+1$. The descendants of the white reading occupy the emptied A1 and are cut; White's A3, a suicide in every surviving element, stands on the display. So $M(s') \subsetneq \bigcap_{b \in \rho(s')} \mathbb{M}(b)$, and the radius of k -*DSGo* is exactly k .

The proof is now complete. For extra peace of mind on the 3×3 board the theorem is computer verified by exhaustion, if setting $k = 1, 2$ for k -*DSGo*. $\square$

5.2. Chess and faithfulness. We now come to the main reason for studying faithfulness: it gives rise to a qualitative mathematical difference between chess and *Go*.

Theorem 5.3. *$\widetilde{\text{chess}}$ has no simple simultaneization that is faithful within distance one of classical play: equivalently, none of positive faithfulness radius. In particular, the universal simultaneization $\text{Sim}(\widetilde{\text{chess}})$ is not simple.*

Proof. First a technical lemma.

Lemma 5.4. *Let G_1 be a simultaneization of G_0 , and suppose that $s \in B(G_1)$ is obtained from the initial state by a game history consisting of mirror pairs, that is, of simultaneous pairs of the form $(x_i, \mathcal{A}(x_i))$. Then $\mathcal{R}(s) = s$, and consequently $\mathcal{A}(M(s)) = M(s)$.*

Proof. The mirror of the pair $(x, \mathcal{A}(x))$ is the pair $(\mathcal{A}(\mathcal{A}(x)), \mathcal{A}(x)) = (x, \mathcal{A}(x))$ itself. So applying the symmetry property of Definition 4.2, and the order invariance of E (the last property of Definition 4.1), along the game history, together with $\mathcal{R}(q_0) = q_0$, we get $\mathcal{R}(s) = s$. Then for any m , $E_{G_1}(s, m)$ is defined if and only if $E_{G_1}(\mathcal{R}(s), \mathcal{A}(m)) = E_{G_1}(s, \mathcal{A}(m))$ is defined, giving $\mathcal{A}(M(s)) = M(s)$. $\square$

Now return to the proof of the theorem. Let G_1 be any simultaneization of $\widetilde{\text{chess}}$ faithful within distance one of classical play, with P_0 the White player, and consider the game history of G_1 consisting of the mirror pairs

$$a4/a5; \quad Ra3/Ra6; \quad Rb3/Rb6,$$

in standard chess algebraic notation. The moves of each pair do not interact, so both serializations are pseudo legal with the same outcome, and each state along

this history is semi-classical. By the semi-classicality condition of Definition 4.16 together with faithfulness at the semi-classical states, applied inductively along the history, the history is legal in G_1 and for the resulting state $s^* \in B(G_1)$ we have $\rho(s^*) = \{a^*\}$, with a^* the classical position of Figure 1. Note that this pins down $\rho(s^*)$ for every simultaneization G_1 , whatever its branch reduction mechanism.

Now let m_0 be the White rook move $b3 \rightarrow b5$, and $m_1 = \mathcal{A}(m_0)$ the Black rook move $b6 \rightarrow b4$. Each order of these moves blocks the other: after $b3 \rightarrow b5$ the Black rook's path is blocked at $b5$, while after $b6 \rightarrow b4$ the White rook's path is blocked at $b4$. So in $\text{Sim}(\widetilde{\text{chess}})$ this pair resolves by the lowest priority case of the serialization map P , and

$$P(a^*, m_0, m_1) = \{a_0, a_1\},$$

where in a_0 the White rook stands on $b5$ with the Black rook on $b6$, and a_1 is the mirror image of a_0 . By faithfulness at the semi-classical s^* , the pair (m_0, m_1) , being available in $\text{Sim}(\widetilde{\text{chess}})$ at $\{a^*\}$, is available at s^* ; so $s_1^* = E_{G_1}(s^*, m_0, m_1)$ is defined, and by condition 3 together with faithfulness at s_1^* – a state one turn from s^* – $\rho(s_1^*)$ is a non-empty subset of $\{a_0, a_1\}$. Moreover s_1^* is not final: it is within distance one of the semi-classical s^* , so the finality clause of faithfulness applies, and neither a_0 nor a_1 is final in $\widetilde{\text{chess}}$.

Suppose that $\rho(s_1^*)$ is a singleton, say $\{a_0\}$. The history of s_1^* consists of mirror pairs, so by Lemma 5.4 $M(s_1^*)$ is $\mathcal{A}$ invariant. But by faithfulness at s_1^* , $M(s_1^*) = \mathbb{M}(a_0)$, which is not $\mathcal{A}$ invariant: in a_0 White has rook moves from $b5$, while Black has no rook moves from $b4$. The case of $\{a_1\}$ is mirrored. Thus this branch reduction is forbidden by the symmetry of G_1 , and $\rho(s_1^*) = \{a_0, a_1\}$.

By faithfulness at s_1^* , $M(s_1^*) = \mathbb{M}(a_0) \cap \mathbb{M}(a_1)$, which is computed directly: for each player it consists of the pass move, the moves of the c through h pawns, and the four knight moves. There are no queen side rook moves, as these rooks stand on different squares in a_0 and a_1 , and no a or b pawn moves. Now suppose that G_1 is simple, so that there is a legally obtainable position c of $\widetilde{\text{chess}}$ with $\mathbb{M}(c)$ equal to this set. The interface $\mathbb{M}(c)$ forces every piece of c other than the two queen side rooks, the two a pawns and the two b pawns to stand on its starting square. One now checks the finitely many placements, or absences by capture, of the six remaining pieces, and each fails. For example, if the White b pawn is at $b2$, then $b3$ must be occupied, necessarily by a queen side rook or the Black b pawn, giving respectively a legal rook move from $b3$ and the legal capture $b3 \times c2$, neither of which is in the set; while if the White b pawn is absent, having been captured, and $b2$ is unoccupied, then the $c1$ bishop has the legal move to $b2$, again not in the set. So we have a contradiction, and G_1 is not simple. $\square$

5.3. Mirror-tie chess. The faithfulness hypothesis in Theorem 5.3 cannot be dropped. Define the *mirror-tie game* G_1 : play proceeds as in $\widetilde{\text{chess}}$ but with simultaneously declared moves, as long as the two moves of each turn do not genuinely interact – both orders of play legal and with the same outcome. Thus each state s so reached is semi-classical, with $\rho(s)$ the singleton of its classical position. At the first genuinely conflicting move pair the game locks: only the pass moves remain available. After two consecutive passes the game ends in a tie.

The conditions of Definition 4.16 are readily verified; the locked states, having only passes available, are exempt from the simplicity condition, and at every other

state the interface is that of a classical position. So G_1 is a simple simultaneization of *chess*, of faithfulness radius zero. (Fully degenerate examples also exist: the game in which only passes are ever available is a simple simultaneization of any G_0 with a pass move, vacuously.) Together with Theorem 5.3: the absolute radius of *chess* is zero – its conflicts can be refused, but not played out – while the absolute radius of *Go* is infinite, by Theorem 5.2. Nothing in the mirror-tie construction is special to chess, so every game of our standing kind has absolute radius at least zero. Within the ladder of *Go*: *SGo* continues past every conflict at radius zero, embracing the divergences; each turn of delay buys one rung; *DSGo*, all delay, is faithful everywhere.

Question 5.5. *Which values does the absolute radius attain? Chess and Go realize the extremes: zero and infinity. We do not know a game of finite positive absolute radius, nor whether one exists.*

6. INFORMED PLAY AND CONCLUDING REMARKS

The focus of this section is our informed play experiments: their methodology and partial results, for both games. All figures below are for the 5×5 board, and may be compared directly with the random play columns of the table of Section 2.1.

6.1. Equilibrium guided play. Our equilibrium guided play is computed as follows: at each turn both players are offered a random menu of five moves; the resulting 5×5 matrix of joint outcomes is filled by averaging the final margins of short uniformly random continuations;¹⁰ and each player samples their move from an optimal mixed strategy of this zero sum matrix game. Three technical findings. First, at about 74% of the turns the estimated one turn game has *no pure saddle point*, so the players must randomize – consistent with Section 4.2, where mixing is proved necessary already on the 3×3 board (re-estimating the matrices at fivefold rollout depth, on fixed positions and menus, lowers the mixing fraction from 74% to 54% on a control sample; genuine mixing remains). Second, the within turn value spread is large, about 16 points of the twenty five: the joint move choice of a single turn swings most of the board. Third, the summary table of the next subsection shows that equilibrium guided play leaves the creation of entanglement essentially unchanged relative to the random play columns of the table of Section 2.1 (3.2 q-stones created per game against 3.3) and resolves slightly more of it by the end (final branch count 3.2 against 3.4): the entanglement is a stable feature of the game, of order one under random and informed play alike.

6.2. Versus informed play in classical *Go*. For the classical game we use greedy rollout guided play: at each move the player evaluates up to six candidate moves, and the pass, by twelve random rollouts each (the convention of the preceding footnote), and plays the best. The summary of informed self play, for both games, in the format of the table of Section 2.1 (162 and 165 games; unbounded play in both cases):

¹⁰Rollouts use the standard Monte Carlo payout convention: uniformly random moves, except that a player does not fill their own single point eyes and does not play suicide, passing when no other move is available.

	<i>SGo</i>	classical <i>Go</i>
mean margin	11.7	11.4
draws	5.6%	6.7%
games ending on their own	100%	100%
mean length in turns	21.2	17.8
q-stones created per game	3.2	–
q-stones on the final display	2.3	–
peak number of branches	4.2	–
final number of branches	3.2	–

For *SGo* the table is close to its random play columns: the margin eases (11.7 against 13.6), the draw rate and game length barely move, and the entanglement is unchanged in creation and slightly more resolved by the end. For the classical game the change is termination: informed play supplies the endings that random play never reaches – every game now ends on its own, in 17.8 turns on average – while *SGo* terminates by rule either way.

At this shallow search depth no measurable first mover advantage emerges in *Go* (signed mean margins statistically zero). Hence we are not correcting by any Komi, which is also why draws appear for *Go* in the table above. We note that *Go* is solved on the 5×5 board – a win for Black by 25 points under standard rule formalizations [25] – a reminder that these are partial, shallow depth results.

Whether the above observations persist for equilibrium guided play on large boards, along with the sharpness observed in Section 2.1, and to what extent the display determines the entanglement state (Question 6.1), we leave to future study.

6.3. Does the display determine the entanglement?

Question 6.1. *To what extent is the entanglement state of a legal *SGo* history determined by its sequence of displays? We do not know. About one in nine quantum-free displays in our samples carried more than one branch, so it is not determined outright. Yet in about a hundred thousand sampled coincidences of display and time stamps, on the 4×4 board, the two entanglement states were always comparable: one contained in the other, the smaller a branch reduction of the larger, and so consistent with either history.*

6.4. Lazy evaluation. Finally, we note that the pattern of resolution in *SGo* – branchings accumulate silently until objective information forces their evaluation – is also familiar in computer science, notably in lazy evaluation [8, 11], where a deferred computation is forced exactly when its value is demanded; we thank a referee for this observation.

REFERENCES

- [1] R. ALUR, T. A. HENZINGER, AND O. KUPFERMAN, *Alternating-time temporal logic*, Journal of the ACM, 49 (2002), pp. 672–713. 4
- [2] E. R. BERLEKAMP AND D. WOLFE, *Mathematical Go: Chilling Gets the Last Point*, AK Peters, Natick, MA, 1994. 1
- [3] A. CINCOTTI AND H. IIDA, *The game of synchronized cutcake*, in IEEE Symposium on Computational Intelligence and Games (CIG 2007), IEEE, 2007, pp. 374–379. 1
- [4] J. H. CONWAY, *On Numbers and Games*, Academic Press, 1976. 1, 4.3
- [5] L. DOYEN AND J.-F. RASKIN, *Games with imperfect information: theory and algorithms*, in Lectures in Game Theory for Computer Scientists, Cambridge University Press, 2011. 4.12
- [6] J. EISERT, M. WILKENS, AND M. LEWENSTEIN, *Quantum games and quantum strategies*, Physical Review Letters, 83 (1999), pp. 3077–3080. 1

- [7] D. GALE, H. W. KUHN, AND A. W. TUCKER, *On symmetric games*, in Contributions to the Theory of Games, Vol. I, vol. 24 of Annals of Mathematics Studies, Princeton University Press, 1950. 4.2
- [8] P. HENDERSON AND J. H. MORRIS, JR., *A lazy evaluator*, in Proceedings of the 3rd ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL '76), ACM, 1976, pp. 95–103. 6.4
- [9] M. HUGGAN, R. J. NOWAKOWSKI, AND P. OTTAWAY, *Simultaneous combinatorial game theory*. arXiv:1810.02870, 2018. 1, 4
- [10] M. A. HUGGAN, *Studies in Alternating and Simultaneous Combinatorial Game Theory*, ph.d. thesis, Dalhousie University, Halifax, Nova Scotia, Canada, 2019. 1
- [11] J. HUGHES, *Why functional programming matters*, The Computer Journal, 32 (1989), pp. 98–107. 6.4
- [12] S. KAKUTANI, *A generalization of Brouwer's fixed point theorem*, Duke Mathematical Journal, 8 (1941), pp. 457–459.
- [13] J. F. J. NASH, *Equilibrium points in n -person games*, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49. 1, 4.2
- [14] M. J. OSBORNE AND A. RUBINSTEIN, *A Course in Game Theory*, MIT Press, 1994. 4.2
- [15] R. PENROSE, *The Emperor's New Mind*, Oxford University Press, 1989. 1
- [16] M. O. RABIN AND D. SCOTT, *Finite automata and their decision problems*, IBM Journal of Research and Development, 3 (1959), pp. 114–125. 4.12
- [17] A. RANCHIN, *Quantum Go*. arXiv:1603.04751, 2016. 1
- [18] J. H. REIF, *The complexity of two-player games of incomplete information*, Journal of Computer and System Sciences, 29 (1984), pp. 274–301. 4.12
- [19] J. M. ROBSON, *The complexity of go*, Proceedings of the International Congress of Mathematicians, (1983), pp. 661–666. 1
- [20] U. SCHWALBE AND P. WALKER, *Zermelo and the early history of game theory*, Games and Economic Behavior, 34 (2001), pp. 123–137. 4.2
- [21] L. S. SHAPLEY, *Stochastic games*, Proceedings of the National Academy of Sciences, 39 (1953), pp. 1095–1100. 4
- [22] M. THIELSCHER, *The general game playing description language is universal*, in Proceedings of the 22nd International Joint Conference on Artificial Intelligence (IJCAI 2011), AAAI Press, 2011. 4
- [23] J. TROMP AND G. FARNEBÄCK, *Combinatorics of Go*, in Computers and Games (CG 2006), vol. 4630 of Lecture Notes in Computer Science, Springer, 2007, pp. 84–99. 5.1
- [24] J. TROMP AND B. TAYLOR, *The logical rules of Go*. <https://tromp.github.io/go.html>, 1995. 5.1
- [25] E. C. D. VAN DER WERF AND M. H. M. WINANDS, *Solving Go for rectangular boards*, ICGA Journal, 30 (2009), pp. 77–88. 6.2